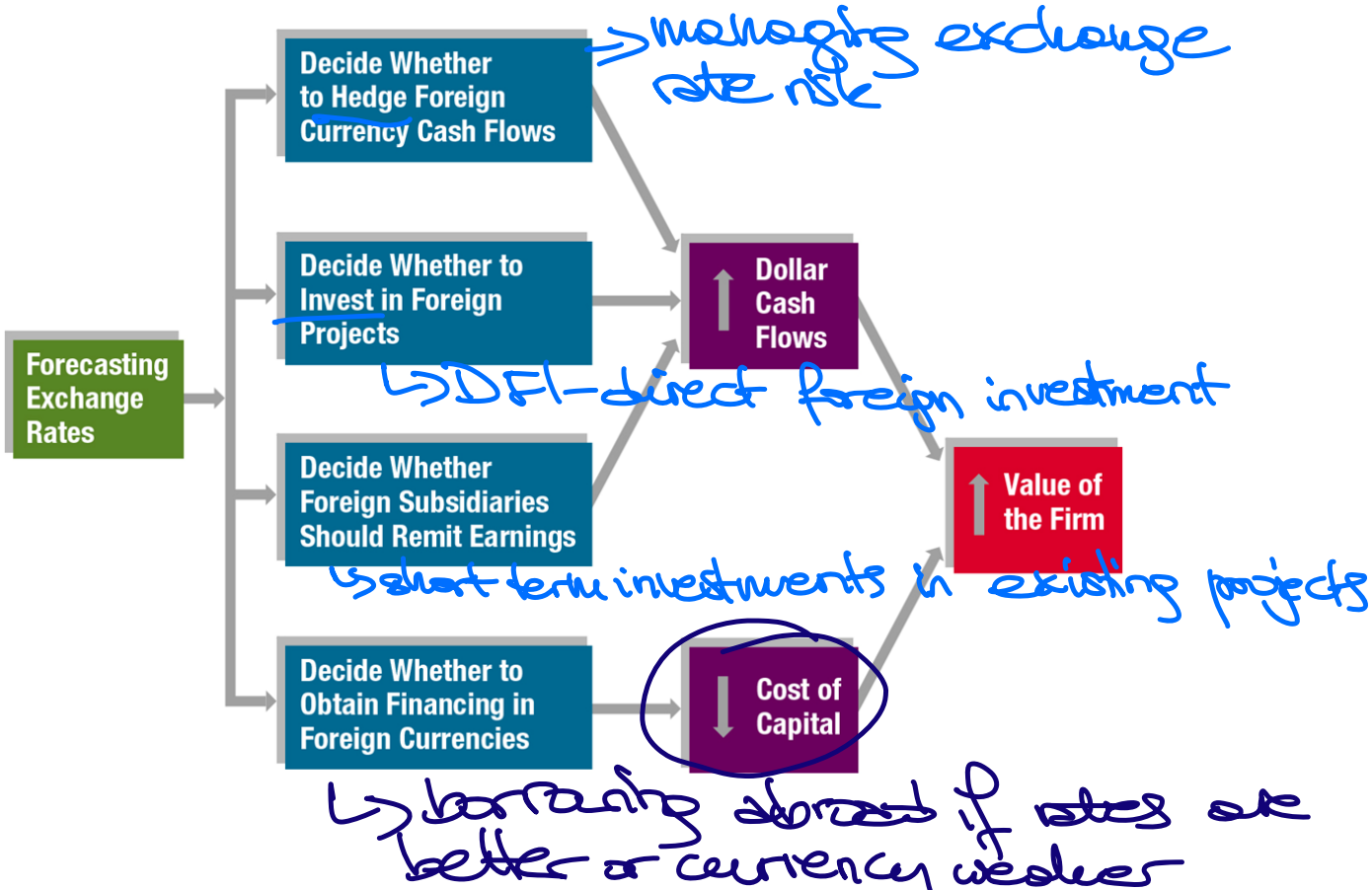
Ch. 9 Forecasting

Exchange Rates

Why Firms Forecast Exchange Rates



Forecasting FX considering Direct Foreign Investment (DFI)

→ local currency is strong, foreign currency is weak

→ if foreign currency is currently weak and expected to appreciate in the long term
→ GOOD INESTING OPPORTUNITY

→ if foreign currency is ~~strong~~ and expected to depreciate, BAD INVESTING OPPORTUNITY

⇒ Lower cost of capital increases valuation.

$$V_0 = \frac{\text{Cash Flow}_1}{(1 + \text{cost of Capital}_r)^1}$$

CF = 1,000,000

a) $r = 10\%$ > b) $r = 2\%$

$V_0 = \frac{1,000,000}{1.1} =$ < $V_0 = \frac{1,000,000}{1.02} =$

$= 909,090.91$ $= 980392.16$

Forecasting Techniques

- Technical Forecasting
- Fundamental Forecasting
- Market-Based Forecasting
- Mixed Forecasting

Technical Forecasting

- Use of past data to infer patterns of movement, not based in economic analysis
- comparing moving averages of return between periods → momentum (bullish vs bearish)
  - upward pattern
  - downward pattern
- comparing volatilities

Fundamental Forecasting

- using economic variables to infer movements in exchange rates

$$\%\Delta FX = f(\Delta INT, \Delta INFL, \Delta INC, \Delta GC, \Delta EXP)$$

GC: government controls

→ time series regression

→ Purchasing Power Parity, IFE

Purchasing Power Parity

$$\%\Delta FX = Infl_h - Infl_f$$

US Inflation is 1% and Chinese inflation is 6%

Current $S_0 = \$0.1429/RMB$

$\%\Delta FX = 1\% - 6\% = -5\%$ → Chinese currency will weaken by 5%

$$E(S_1) = S_0 \cdot (1 + \%\Delta FX)$$

$$= \$0.1429 \cdot (1 - 5\%)$$

$$= \$0.1429 \cdot 0.95 = \$0.1357/RMB$$

Regression Analysis

$$y_{\%\Delta FX} = a + b\,INFL + c\,INT + d\,INC + e\,GC + f\,EXP + error$$

$y = 0.5$

→ lagging variables – regress change in exchange rate today on change in interest the month before

Market-Based Forecasting

→ inferring changes rates based on market rates

→ using spot market → rates in strong-form efficient markets

$E(S_1) = S_0$

$= E(\Delta FX) = 0$

→ using forward market → forward premium is used as estimator of future exchange rates

$E(\%\Delta FX) = \frac{F}{S} - 1 = p$

$F = S \cdot (1+p)$

$\frac{F}{S} - 1 = p$

Mixed Forecasting

- using combination of forecasting methods
- calculating E(ΔFX) using multiple methods:
  - using median result
  - using weighted average
  - comparing over different time spans (short-term or long-term)

Forecasts of the Mexican Peso Drawn from Each Forecasting Technique

FORECAST TECHNIQUE	FACTORS CONSIDERED	SITUATION	FORECAST
Technical	Recent movement in peso	The peso's value declined below a specific threshold level in the last few weeks.	The peso's value will continue to fall now that it is beyond the threshold level.
Fundamental	Economic growth, inflation, interest rates	Mexico's interest rates are high, and inflation should remain low.	The peso's value will rise as U.S. investors capitalize on the high interest rates by investing in Mexican securities.
Market-based	Spot rate, forward rate	The peso's forward rate exhibits a significant discount, which is attributed to Mexico's relatively high interest rates.	Based on the forward rate, which provides a forecast of the future spot rate, the peso's value will decline.

Assessment of Forecast Performance

- Forecast Error among techniques

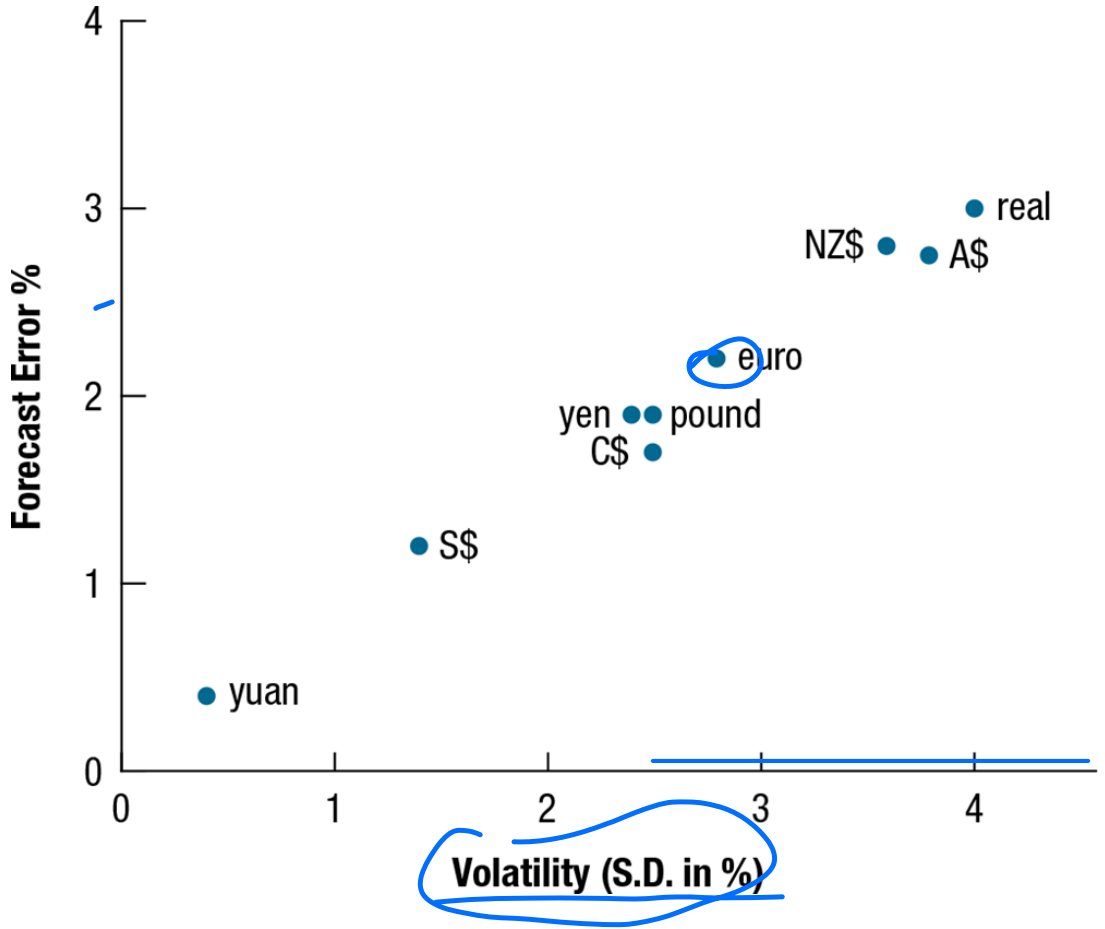
$$\left.\begin{array}{l}\text{Absolute forecast error as a percentage} \\ \text{of the realized value}\end{array}\right\} = \frac{\left|\text{Forecasted value} - \text{Realized value}\right|}{\text{Realized value}}$$

- Forecast bias over time

Comparing Forecast Errors among Forecast Techniques

(1)	(2)	(3)	(4)	(5)	(6)	(7) = (5) − (6)
PERIOD	PREDICTED VALUE OF ZLOTY BY MODEL 1	PREDICTED VALUE OF ZLOTY BY MODEL 2	REALIZED VALUE OF ZLOTY	ABSOLUTE FORECAST ERROR USING MODEL 1	ABSOLUTE FORECAST ERROR USING MODEL 2	DIFFERENCE IN ABSOLUTE FORECAST ERRORS (MODEL 1 − MODEL 2)
1	$.20	$.24	$.16	$.04	$ .08	$−.04
2	.18	.20	.14	.04	.06	−.02
3	.24	.20	.16	.08	.04	.04
4	.26	.20	.22	.04	.02	.02
5	.30	.18	.28	.02	.10	−.08
6	.22	.32	.26	.04	.06	−.02
7	.16	.20	.14	.02	.06	−.04
8	.14	.24	.10	.04	.14	−.10
				Sum = .32 Mean = .04	Sum = .56 Mean = .07	Sum = −.24 Mean = −.03

How the Forecast Error Is Affected by Volatility



Evaluating Bias

- Graphical analysis
- Regression

